

Quantum Mechanics – Homework 2 Solutions

Lectures 4–5

Instructor version. The main calculation and the physical meaning of each result are included.

Problem 1. Normalization and the Born Rule (20 points)

(a) A normalized wave function satisfies

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1.$$

Since $|\psi(x)|^2 = A^2 e^{-2\kappa|x|}$ is even,

$$1 = 2A^2 \int_0^{\infty} e^{-2\kappa x} dx.$$

(b) Evaluating the integral gives

$$1 = 2A^2 \left(\frac{1}{2\kappa} \right) = \frac{A^2}{\kappa}.$$

Since A is positive,

$$\boxed{A = \sqrt{\kappa}}.$$

(c) The probability is

$$\begin{aligned} P\left(-\frac{1}{\kappa} < x < \frac{1}{\kappa}\right) &= 2 \int_0^{1/\kappa} \kappa e^{-2\kappa x} dx \\ &= 2\kappa \left[-\frac{e^{-2\kappa x}}{2\kappa} \right]_0^{1/\kappa} \\ &= \boxed{1 - e^{-2}} \\ &\simeq \boxed{0.865}. \end{aligned}$$

Thus about 86.5% of position measurements lie in this interval.

(d) In one dimension, $|\psi(x)|^2 dx$ must be dimensionless. Therefore

$$[\psi] = [A] = L^{-1/2}.$$

This also follows from $A = \sqrt{\kappa}$, since $[\kappa] = L^{-1}$. The value $|\psi(x)|^2$ has dimensions L^{-1} , so it is a probability per unit length. A probability is obtained only after it is integrated over an interval.

Problem 2. Normalization of a Superposition (20 points)

- (a) For either
- $n = 1$
- or
- $n = 2$
- ,

$$\int_0^L \phi_n^2(x) dx = \frac{2}{L} \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = 1.$$

To calculate the overlap, use $2 \sin a \sin b = \cos(a - b) - \cos(a + b)$:

$$\begin{aligned} \int_0^L \phi_1(x) \phi_2(x) dx &= \frac{1}{L} \int_0^L \left[\cos\left(\frac{\pi x}{L}\right) - \cos\left(\frac{3\pi x}{L}\right) \right] dx \\ &= \boxed{0}. \end{aligned}$$

The two functions are normalized and orthogonal.

- (b) The norm of the superposition is

$$\int_0^L |\psi_\alpha|^2 dx = N^2 \int_0^L (\phi_1^2 + \phi_2^2 + e^{i\alpha} \phi_1 \phi_2 + e^{-i\alpha} \phi_1 \phi_2) dx.$$

The two cross-term integrals vanish by orthogonality. Hence

$$1 = N^2(1 + 1) = 2N^2, \quad \boxed{N = \frac{1}{\sqrt{2}}}.$$

The phase appears only in the cross terms, so it does not affect the total norm when the component states are orthogonal.

- (c) Using
- $N = 1/\sqrt{2}$
- ,

$$\begin{aligned} |\psi_\alpha(x)|^2 &= \frac{1}{2} (\phi_1 + e^{i\alpha} \phi_2)^* (\phi_1 + e^{i\alpha} \phi_2) \\ &= \boxed{\frac{1}{2} [\phi_1^2(x) + \phi_2^2(x) + 2 \cos \alpha \phi_1(x) \phi_2(x)]}. \end{aligned}$$

The term $2 \cos \alpha \phi_1 \phi_2$ is the interference term.

- (d) Each component contributes one half of its total probability to the left half of the interval:

$$\int_0^{L/2} \phi_1^2 dx = \int_0^{L/2} \phi_2^2 dx = \frac{1}{2}.$$

The overlap restricted to the left half is not zero:

$$\begin{aligned}\int_0^{L/2} \phi_1 \phi_2 dx &= \frac{2}{\pi} \int_0^{\pi/2} \sin u \sin(2u) du \\ &= \frac{2}{\pi} \int_0^{\pi/2} 2 \sin^2 u \cos u du \\ &= \frac{4}{3\pi}.\end{aligned}$$

Therefore

$$\begin{aligned}P(0 < x < L/2) &= \frac{1}{2} \left[\frac{1}{2} + \frac{1}{2} + 2 \cos \alpha \left(\frac{4}{3\pi} \right) \right] \\ &= \boxed{\frac{1}{2} + \frac{4 \cos \alpha}{3\pi}}.\end{aligned}$$

The three requested values are

α	$P(0 < x < L/2)$
0	$\frac{1}{2} + \frac{4}{3\pi} \simeq 0.924$
$\frac{\pi}{2}$	$\frac{1}{2}$
π	$\frac{1}{2} - \frac{4}{3\pi} \simeq 0.0756$

The total normalization is independent of the relative phase, but the spatial probability distribution is not.

Problem 3. Position Statistics (20 points)

(a) Normalization requires $y = x/L$, so that $x = Ly$ and $dx = L dy$. Therefore

$$\begin{aligned}1 &= A^2 \int_0^L x^2 (L-x)^4 dx \\ &= A^2 L^7 \int_0^1 y^2 (1-y)^4 dy \\ &= A^2 \frac{L^7}{105}.\end{aligned}$$

Thus

$$\boxed{A = \frac{\sqrt{105}}{L^{7/2}}}.$$

(b) The probability in the left half is

$$\begin{aligned}
 P(0 < x < L/2) &= A^2 \int_0^{L/2} x^2 (L - x)^4 dx \\
 &= 105 \int_0^{1/2} y^2 (1 - y)^4 dy \\
 &= 105 \left[\frac{y^3}{3} - y^4 + \frac{6y^5}{5} - \frac{2y^6}{3} + \frac{y^7}{7} \right]_0^{1/2} \\
 &= \boxed{\frac{693}{896}} \\
 &\simeq \boxed{0.773}.
 \end{aligned}$$

(c) The first moment is

$$\begin{aligned}
 \langle x \rangle &= A^2 \int_0^L x^3 (L - x)^4 dx \\
 &= 105L \int_0^1 y^3 (1 - y)^4 dy \\
 &= 105L \left(\frac{1}{280} \right) = \boxed{\frac{3L}{8}}.
 \end{aligned}$$

The second moment is

$$\begin{aligned}
 \langle x^2 \rangle &= A^2 \int_0^L x^4 (L - x)^4 dx \\
 &= 105L^2 \int_0^1 y^4 (1 - y)^4 dy \\
 &= 105L^2 \left(\frac{1}{630} \right) = \boxed{\frac{L^2}{6}}.
 \end{aligned}$$

(d) The variance is

$$\begin{aligned}
 (\Delta x)^2 &= \frac{L^2}{6} - \left(\frac{3L}{8} \right)^2 \\
 &= \frac{5L^2}{192}.
 \end{aligned}$$

Therefore

$$\boxed{\Delta x = L \sqrt{\frac{5}{192}} = \frac{\sqrt{15}}{24} L}.$$

(e) Inside the interval, the probability density is proportional to $x^2(L - x)^4$. For $0 <$

$x < L$, its logarithmic derivative is

$$\frac{d}{dx} \ln[x^2(L-x)^4] = \frac{2}{x} - \frac{4}{L-x}.$$

Setting this to zero gives

$$2(L-x) = 4x, \quad \boxed{x_{\text{most}} = \frac{L}{3}}.$$

This is not equal to $\langle x \rangle = 3L/8$. The most probable value is the location where the probability density is largest. The expectation value is the probability-weighted average of many position measurements.

Problem 4. The Momentum Operator (20 points)

Let $k_0 = \pi/L$ and $C = \sqrt{2/L}$, so that $\psi_q = C \sin(k_0 x) e^{iqx}$.

(a) Since $|e^{iqx}|^2 = 1$,

$$|\psi_q(x)|^2 = C^2 \sin^2(k_0 x).$$

The phase factor changes neither the position probability density nor its integral. The state therefore remains normalized.

(b) Applying the momentum operator gives

$$\begin{aligned} \hat{p}\psi_q &= -i\hbar C \frac{d}{dx} [\sin(k_0 x) e^{iqx}] \\ &= \hbar q \psi_q - i\hbar C k_0 \cos(k_0 x) e^{iqx}. \end{aligned}$$

Thus

$$\boxed{\hat{p}\psi_q = \hbar q \psi_q - i\hbar \sqrt{\frac{2}{L}} \frac{\pi}{L} \cos\left(\frac{\pi x}{L}\right) e^{iqx}}.$$

The result is not a constant times ψ_q , because it contains a cosine term. Therefore ψ_q is not a momentum eigenfunction.

(c) The expectation value is

$$\begin{aligned} \langle p \rangle &= \int_0^L \psi_q^* \hat{p} \psi_q dx \\ &= \hbar q \int_0^L |\psi_q|^2 dx - i\hbar C^2 k_0 \int_0^L \sin(k_0 x) \cos(k_0 x) dx. \end{aligned}$$

The first integral is one. The second integral vanishes because

$$\int_0^L \sin(k_0 x) \cos(k_0 x) dx = \left[\frac{\sin^2(k_0 x)}{2k_0} \right]_0^L = 0.$$

Hence

$$\boxed{\langle p \rangle = \hbar q}.$$

(d) Applying $\hat{p}$ twice gives

$$\hat{p}^2 \psi_q = \hbar^2 (q^2 + k_0^2) \psi_q - 2i\hbar^2 q C k_0 \cos(k_0 x) e^{iqx}.$$

The cosine cross term again integrates to zero. Therefore

$$\boxed{\langle p^2 \rangle = \hbar^2 \left(q^2 + \frac{\pi^2}{L^2} \right)}.$$

The variance and uncertainty are

$$(\Delta p)^2 = \langle p^2 \rangle - \langle p \rangle^2 = \frac{\pi^2 \hbar^2}{L^2},$$

$$\boxed{\Delta p} = \boxed{\frac{\pi \hbar}{L}}.$$

(e) Increasing q shifts the mean momentum according to $\langle p \rangle = \hbar q$. The momentum uncertainty $\Delta p = \pi \hbar / L$ is unchanged. The phase e^{iqx} shifts the centre of the momentum distribution without changing its width.

Problem 5. Momentum-Space Statistics (20 points)

(a) The two norms are

$$\int_0^L |u_+(x)|^2 dx = \int_0^L |u_-(x)|^2 dx = \frac{1}{L} \int_0^L dx = 1.$$

Their overlap is

$$\begin{aligned} \int_0^L u_+^*(x) u_-(x) dx &= \frac{1}{L} \int_0^L e^{-2ikx} dx \\ &= \frac{e^{-2ikL} - 1}{-2ikL} = 0, \end{aligned}$$

because $kL = 2\pi$. Therefore

$$\int_0^L |\psi|^2 dx = \left| \frac{\sqrt{3}}{2} \right|^2 + \left| \frac{i}{2} \right|^2 = \frac{3}{4} + \frac{1}{4} = 1.$$

(b) Applying the momentum operator gives

$$\hat{p}u_+ = \hbar k u_+, \quad \hat{p}u_- = -\hbar k u_-.$$

Thus the possible momentum results and probabilities are

momentum result	probability
$+\hbar k$	$3/4$
$-\hbar k$	$1/4$

(c) The momentum expectation value is

$$\langle p \rangle = \frac{3}{4}(\hbar k) + \frac{1}{4}(-\hbar k) = \boxed{\frac{\hbar k}{2}}.$$

Both possible values have the same square, so

$$\langle p^2 \rangle = \frac{3}{4}(\hbar k)^2 + \frac{1}{4}(-\hbar k)^2 = \boxed{\hbar^2 k^2}.$$

Hence

$$(\Delta p)^2 = \hbar^2 k^2 - \left(\frac{\hbar k}{2} \right)^2 = \frac{3}{4} \hbar^2 k^2,$$

$$\boxed{\Delta p} = \boxed{\frac{\sqrt{3}}{2} \hbar k}.$$

(d) Direct multiplication gives

$$|\psi(x)|^2 = \frac{1}{L} \left| \frac{\sqrt{3}}{2} e^{ikx} + \frac{i}{2} e^{-ikx} \right|^2$$

$$= \boxed{\frac{1}{L} \left[1 + \frac{\sqrt{3}}{2} \sin(2kx) \right]}.$$

Each plane wave separately has a uniform probability density. Their superposition contains cross terms whose phase varies with position. These interference terms produce the position-dependent density.